

- This exam contains 7 pages (including the front and back of this cover page) and 11 problems. Check to see if any pages are missing.
- This is the second exam and will count for 20% of your final grade. You have until Monday, April 25th at 11:59 PM to complete this exam. You may use consult your notes and textbook during this exam, but no other outside sources may be consulted.

The following rules apply:

- Every problem on this exam, other than true-or-false problems, requires work of some form. An incorrect answer supported by substantially correct calculations and explanations will receive partial credit. **A correct answer that has no supporting work will not receive full credit.**
- Organize your work, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little to no credit.
- There are 140 points total in this exam, as well as 1 bonus questions worth a total of 10 possible points.

Page:	2	3	4	5	6	7	Total
Points:	20	20	20	40	20	20	140
Score:							

Bonus:	
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1. Mark each statement as ☐ True or ☐ False by completely shading in the box corresponding to your answer.

(a) ☐ 2 points The function $f(z) = \frac{z^2}{\sin z}$ has a removable singularity at $z = 0$.

☐ True☐ False

(b) ☐ 2 points Suppose f is analytic on $\mathbb{C} \setminus \{0\}$ and has residue 1 at $z = 0$. Then, the residue of $(f(z))^2$ at $z = 0$ is also 1.

☐ True☐ False

(c) ☐ 2 points If f is entire and $|f(z)| \leq 1$ for all z , then f is constant.

☐ True☐ False

(d) ☐ 2 points Suppose f_n is a sequence of analytic functions and $f_n \rightarrow f$ uniformly on a region A . Then, $f'_n(z) \rightarrow f'(z)$ uniformly on A .

☐ True☐ False

(e) ☐ 2 points $\text{Res}(f + g; z) = \text{Res}(f; z) + \text{Res}(g; z)$.

☐ True☐ False

2. ☐ 10 points Find the Laurent expansion of $\frac{e^z}{z} + e^{1/z}$ around $z = 0$.

3. 10 points Find the residues at all singularities of $\tan z$.

4. 10 points Let $f(z) = \cos\left(1 - \frac{1}{z}\right)$. Compute $\text{Res}(f; 0)$.

5. Evaluate the following residues:

(a) 5 points $\text{Res} \left(\frac{e^z}{z^3 - 1}; 1 \right)$

(b) 5 points $\text{Res} \left(\frac{1}{z^6 - \omega}; \omega \right)$

(c) 5 points $\text{Res} \left(\tan z; \frac{\pi}{2} \right)$

(d) 5 points $\text{Res} (z^2; 0)$

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6. 20 points Construct a meromorphic function f with a simple pole of residue 1 at all positive integers (i.e. such that $\text{Res}(f; k) = 1$ for $k = 1, 2, 3, \dots$).
7. 20 points Construct a meromorphic function f with a pole of order 2 at $z = 1 + i$ and essential singularities at $z = 0$ and $z = 1$.

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8. 10 points Find the residue at ∞ of $\frac{z-1}{z+1}$.
9. 10 points Prove that if f has a pole of order m at $z = z_0$, then f^2 has a pole of order $2m$ at $z = z_0$.

10. 20 points Compute $\int_{-\pi}^{\pi} \frac{d\theta}{1 + \sin^2(\theta)}$.

11. 10 points (bonus) Show that $\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}$.